Ageing or Attack? Cause-Aware Learning for Safe Self-Calibration Under Disguised Sensor Poisoning

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Abstract

IoT pipelines increasingly recalibrate, re-baseline and retrain on the data they accept, so a false-data injection shaped like benign transducer ageing is absorbed as the new normal and later hides genuine faults. We formulate drift-versus-poisoning attribution as cause-aware learning under adversarial distribution shift, with an attacker hierarchy that matches benign ageing in magnitude and rate, and show that an on-path attacker is observationally equivalent to drift while a physics-anchored, bounded correction still bounds its harm. PIRCAD judges each channel against a leave-one-out physics expectation, classifies the cause (normal, drift, poisoning, process event) from residual-character and constraint-localisation features, and lets the verdict gate every adaptive element. Across 180 held-out simulated episodes, PIRCAD reaches macro-F1 0.943 (random forest) and 0.962 (gradient boosting); a supervised 1D-CNN (0.938) and a generic-feature forest (0.974) are as accurate in distribution. Without retraining, under eleven simulated distribution shifts, PIRCAD's worst case is 0.712 against 0.477 and 0.288. In an attack-then-fault experiment, a downstream safety alarm fired in 100 % of 120 episodes, while tuned Kalman, Page-Hinkley and EWMA self-calibration never raised it. On real Intel Berkeley Lab data with mote-disjoint splits, the data-driven learners lead on labels and the CNN also withstands a sensor-noise shift; we report this and every other limitation, including three defects that the stronger evaluation exposed in our earlier pipeline.

Keywords: Sensor drift; Data poisoning; Self-calibration; Root-cause classification; Physics-informed machine learning; Adversarial robustness; Internet of Things; Fog computing

1. Introduction

Networked sensing underpins industrial monitoring, building automation and environmental observation, and every analytic, control decision and safety interlock computed downstream inherits the trustworthiness of the stream it is given [1,2]. That stream is produced by large populations of cheap, unattended transducers, and it goes bad for two very different reasons. Transducers age: their output acquires a slow, smooth bias, and a drifting sensor is still a working instrument that should be recalibrated rather than discarded [3,4]. And streams are attacked: false-data injection fabricates readings that are individually plausible [2,5], and acoustic or thermal injection at the transducer itself bypasses every cryptographic protection on the link [6–8].

The two causes call for opposite responses, yet they can be made to look identical. A bias that grows by a few hundredths of a degree per minute is exactly what an ageing thermistor produces and exactly what a patient attacker produces. This matters because modern IoT pipelines *adapt*: they recalibrate against their own accepted data [3], re-estimate baselines and thresholds [9], and detect and absorb concept drift to suppress false alarms [10,11]. An adaptive element that treats every slow, persistent deviation as benign drift will absorb an injection shaped like drift into its own reference, and the damage outlasts the attack: once the reference has moved, genuine faults are under-reported. This has been shown against drift-adaptive detectors in industrial control [12,13] and, physically, by acoustically

poisoning the training data of an on-device anomaly detector until the machine fault it existed to detect became invisible [7].

The dominant machine-learning formulation does not pose the question that would prevent this. Time-series anomaly detectors emit a score or a flag [14–16]; drift detectors emit a change point [11]. Neither carries a vocabulary of *causes*, so a system built on them has nothing on which to condition its response, and it does the same thing whatever the reason for the deviation. Recent work has begun to separate faults from attacks [5,17–19] and adversarial from real concept drift [20], but only between hypotheses that differ in their signature at the plant, not against an injection deliberately matched to benign ageing; and none of it closes the loop from the verdict to *which adaptive elements may learn from the data*.

This paper treats the problem as one of **cause-aware learning under adversarial distribution shift**. We ask a fog-layer learner to decide, for every deviation, whether it is benign transducer drift, data poisoning, or a genuine excursion of the monitored process; and to let that verdict gate every element that adapts. The approach, PIRCAD (physics-anchored root-cause attribution with adaptation governance), rests on one structural idea: each channel is judged against an expectation that it cannot influence. A leave-one-channel-out (dedicated-observer) estimator anchored on the other instruments and on the process physics produces a residual in which a slow bias appears at full amplitude, and whose *character* — not its size or rate — differs between a drifting transducer that still observes the real process and a substituted stream that does not.

The contributions are:

- **A formulation and threat model** for drift-versus-poisoning attribution in which the adversary matches benign ageing in magnitude and rate, organised as a hierarchy of attacker knowledge. We state the limit of the problem explicitly: an on-path attacker that adds a ramp to the genuine signal is observationally equivalent to drift on the attacked channel (Proposition 1), and we show that a physics-anchored, bounded correction nevertheless bounds the damage such an attacker can do (Proposition 2).
- **A physics-anchored root-cause learner**: leave-one-out residuals, a compact set of residual-character features, and a localisation of physical-constraint violations whose hypothesis signatures come from perturbing the model rather than from labelled faults. The learner is back-end agnostic; we report five.
- **Adaptation governance**: a per-channel state machine that withholds suspect data from every adaptive element, with a cause-dependent response — bounded recalibration for drift, quarantine for poisoning, forward-and-alarm for process events — and a rule that no irreversible action (quarantine) is taken on an uncertain verdict, so that one misclassification cannot lock a serviceable sensor out of correction.
- **A statistically powered evaluation** against supervised deep and feature-based learners, unsupervised detectors and tuned self-calibrating estimators: 30 held-out seeds with paired tests, an attacker-capability sweep, eleven distribution-shift conditions without retraining, an end-to-end safety experiment, and a real-data study on the Intel Berkeley Research Lab deployment with sensor-disjoint train and test splits [21].
- **Reproducibility**: all code, the attack generators and the scripts that produce every number and figure are released; Section 7.2 also documents three defects that the stronger evaluation exposed in our own earlier pipeline, and their fixes.

Section 2 reviews related work. Section 3 formalises the problem and the threat model. Section 4 describes the method, Section 5 the experimental design, and Section 6 the results. Section 7 discusses limitations and threats to validity, and Section 8 concludes.

2. Related work

Anomaly and change detection. Fog-, edge- and on-device anomaly detection place lightweight learners next to the sensors [15,16], and deep time-series anomaly detection has been surveyed comprehensively [14]; work on the decision stage makes thresholds adaptive and uncertainty-aware [9]. Isolation-based scoring [22] and sequential change detection [23] remain the low-cost workhorses. These methods improve the score, not its semantics: the output is a single anomaly label, so there is nothing on which to condition a cause-specific response.

Drift compensation and adaptation. Field recalibration of low-cost sensor networks [3] and learned drift compensation [4] treat slow, persistent, one-directional bias as something to absorb — correctly, when the cause really is ageing — and drift-aware IoT detectors adapt to configuration changes to suppress false positives [10]; concept-drift adaptation more broadly is surveyed in [11]. None of this literature carries an adversarial hypothesis, so absorbing an injection shaped like ageing lies outside its threat model.

Attacks on sensing and on adaptive learners. Stealthy false-data injection against state estimation [24] and replay attacks [25] define the classical adversary; surveys review detectors [2], and random-matrix methods separate injection from replay [5]. Learned detectors are themselves attackable: detectors that predict each sensor from the others can be evaded [26], and gradient-crafted injections are misclassified as legitimate faults [27]. Most directly relevant to this paper, detectors retrained online to follow drift can be poisoned by fake sensor signals so that later attacks go undetected [12,13]; drift detectors can be evaded by adversarially constructed drift [28]; on-device learners have been poisoned physically [7]; and thermal attacks have exploited the temperature-dependent internal calibration of low-cost pollution sensors [8]. Outside IoT, a physical attack on autonomous vehicles first triggers camera–LiDAR online recalibration and then steers the estimate [29] — the same exploit of a self-calibrating element. Poisoning defences for federated learners [30] protect a training corpus, not the calibration offsets and normalisation statistics of a fog pipeline.

Physics-informed and model-based detection. Model-based diagnosis generates residuals against a process model and exploits analytical redundancy [31]; observer-based residuals remain effective for data-efficient diagnosis [32]. Physics-informed learning brings known laws into anomaly detection [33,34], invariant rules among process variables detect attacks in industrial control [35], and digital twins act as observers [36,37]. The nearest edge system, PhySec-Edge, combines process-model residuals, Kalman estimation and cross-sensor checks with learned detectors [38]; it detects attack classes, but benign ageing is not among its hypotheses and the residual is not used to select a response.

Distinguishing faults, attacks and drift. The closest work in objective separates which *kind* of anomaly occurred: attacks from faults in interconnected cyber-physical systems [17], replay attacks from sensor faults [18], malicious anomalies from sensor faults and genuine events with an ensemble pipeline [19], and adversarial from real concept drift in generic data streams [20]. Table 1 positions these against PIRCAD. Three properties are, to our knowledge, not combined elsewhere: an injection matched to benign ageing in *both* magnitude and rate as the decisive hypothesis; attribution grounded in physical constraints whose signatures need no labelled faults; and the verdict gating *every* adaptive element, evaluated by its end effect on a downstream safety function rather than by label accuracy alone.

Table 1. Positioning against the closest prior work. ✓ = addressed; ~ = partially; – = not addressed.

Work	Ageing-matched attack as hypothesis	Drift / attack / process event	Physics-grounded attribution	Cause-dependent response	Gates adaptation	Real data
Drift compensation [3,4]	–	drift only	–	recalibrates	–	✓

Work	Ageing-matched attack as hypothesis	Drift / attack / process event	Physics-grounded attribution	Cause-dependent response	Gates adaptation	Real data
Drift-aware IoT AD [10]	–	–	–	adapts	–	✓
Poisoning of online ICS detectors [12,13]	~ (attack study)	–	–	–	–	✓
Adversarial vs real drift [20]	~	drift vs adversarial	–	–	~	✓
Fault vs attack [17,18]	–	fault vs attack	✓ (observers)	–	–	–
Malicious vs fault vs event [19]	–	✓	–	–	–	✓
PhySec-Edge [38]	–	attack classes	✓	–	–	~
PIRCAD (this work)	**✓ (hierarchy A0–A5)**	**✓**	**✓**	**✓**	**✓ (all elements)**	**✓ (semi-synthetic)**

3. Problem formulation and threat model

3.1 Setting

A fog node receives synchronous samples $y(k) = [y_1(k), \ldots, y_C(k)]$ from C transducers observing a common physical process with state $x(k)$ and known inputs $u(k)$ (actuator commands, or exogenous measurements such as neighbouring sensors). In nominal operation $y_i(k) = h_i(x(k)) + e_i(k)$, with e_i the transducer noise. The node forwards samples to a cloud tier and maintains a set $\mathcal{A}$ of *adaptive elements*: calibration offsets c_i, residual normalisation statistics, the state of its process model, and any learned model that is retrained on accepted data. Each sample of each channel carries a latent cause $z_i(k) \in \{\text{NORMAL, DRIFT, POISON, PROCESS}\}$:

- **DRIFT** — the transducer degrades: $y_i(k) = h_i(x(k)) + e_i(k) + b_i(k)$, with b_i smooth and $|\dot{b}_i| \le \rho_i$, the manufacturer's ageing envelope. The sensor still observes the real process.
- **POISON** — the reported value is replaced or modified by an adversary (Section 3.2).
- **PROCESS** — the plant itself departs from nominal (e.g. an actuator fails); the instruments remain healthy and mutually consistent.

The learning task is to infer $\hat{z}_i(k)$ causally and to choose an action $a_i(k) = \pi(\hat{z}_i(k))$ and an adaptation mask $g_i(k) \in \{0, 1\}$ that decides whether sample k of channel i may update any element of $\mathcal{A}$. The objective is not label accuracy alone but its consequences: drifting sensors should stay in service with their bias removed, poisoned samples should neither be forwarded nor reach $\mathcal{A}$, and process events should be forwarded and alarmed without ever being calibrated away.

3.2 Threat model: an attacker-capability hierarchy

The adversary controls the reported stream of one channel (or, at level A5, several) and wants its injection absorbed as drift. Throughout, the injected deviation is **rate-matched** to benign drift — the same ramp rate, within the ageing envelope — so any detector reasoning about magnitude, persistence or growth rate is at chance by construction. The levels differ in what the attacker knows about the live process:

- **A0 (stale surrogate).** The attacker substitutes a lagged estimate of the process (an exponential moving average with factor 0.02) plus the ramp and white noise of the transducer's variance.
- **A1 (noise-matched).** As A0, but the noise reproduces the transducer's coloured (AR(1)) spectrum.
- **A2 (higher fidelity).** The surrogate tracks the live process with smoothing factor 0.05–0.30.

- **A3 (exact state).** The attacker knows the exact real-time process state (factor 1.0) but not the transducer's noise realisation.
- **A4 (on-path).** The attacker sits on the link and adds the ramp to the *genuine* reading.
- **A5 (coordinated).** One forged temperature drives every instrument the attacker controls (T, P, RH), so the instrument-side physical constraints remain satisfied.

Two further attack families are not rate-matched: isolated spike substitutions and replay of an earlier genuine window. The defender is assumed to have a clean commissioning window and a process model of moderate accuracy, and the attacker is assumed not to control the inputs u(k) at A0–A4.

3.3 What can and cannot be separated

Proposition 1 (observational equivalence). Under A4, with ramp $b(k) = \rho(k - k_0)$ and $|\rho|$ within the ageing envelope, the distribution of the attacked channel's reported sequence is identical to that of the same channel under benign drift with $b_i = b$. *Proof.* Both equal $h_i(x(k)) + e_i(k) + \rho(k - k_0)$ with the same noise realisation; no function of $\{y_i(k)\}$ can separate them, and the other channels are untouched in both cases. ■

Proposition 1 fixes the frontier of the problem: no detector — physics-based or learned — can label A4 correctly from the data alone. What a system *can* control is the consequence of mislabelling it.

Proposition 2 (bounded harm of anchored correction). Let the calibration offset c_i be driven only by residuals against an expectation $\hat{y}_i(k \mid y_{-i}(k), u(k))$ that excludes channel i, with per-sample rate limit δ and total cap c_{max}. If an A4 ramp b(k) is classified as drift, the forwarded corrected value satisfies $|y_i - c_i - h_i(x)| \le |\varepsilon_{ref}| + |e_i| + \max(0, |b| - c_{max}) + L$, where ε_{ref} is the reference error and $L \le \max(0, |\dot{b}| - \delta)\cdot t$ the tracking lag. *Proof.* Because the expectation excludes channel i, the residual estimates $b + \varepsilon_{ref}$; the corrector removes it up to its rate and cap limits. ■

Proposition 2 is the reason the response, and not only the classifier, matters: an equivalent attack that is (unavoidably) called drift is corrected against the physics rather than absorbed into a self-referenced baseline. By contrast, a corrector anchored on the channel's own history — a rolling mean, an EWMA reference, a Page-Hinkley-triggered recalibration — tracks b and leaves the full bias in the forwarded value. Sections 6.2 and 6.4 measure both statements.

4. Method

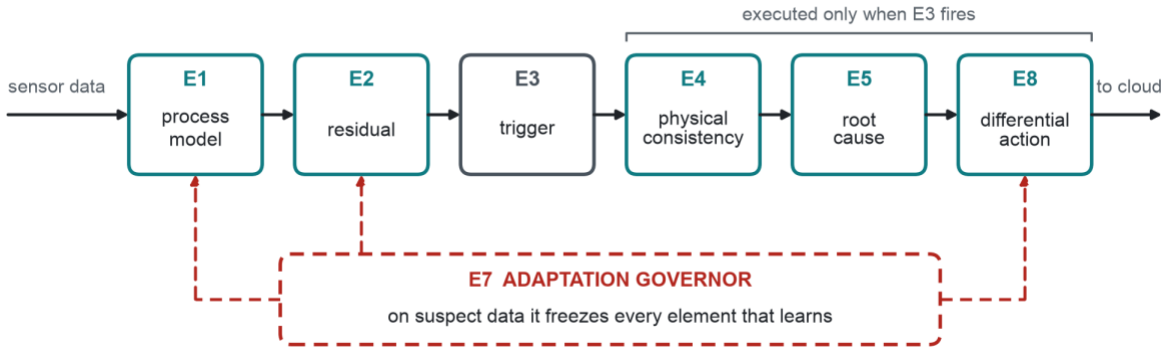


Fig. 1. PIRCAD at the fog node. The expensive stages (E4, E5, E8) run only when the cheap trigger E3 fires. The adaptation governor E7 gates every element that learns: the process model's state, the residual statistics and the calibration offsets.

4.1 Leave-one-out physics residuals (E1–E2)

For each channel i the expected value is produced by a dedicated observer whose state excludes that channel:

$$r_i(k) = y_i(k) - c_i(k) - \hat{y}_i(k \mid y_{-i}(k), u(k)) \quad (1)$$

A single closed-loop estimator that also uses channel i tracks a slow bias on that channel, so its innovation stays at the noise floor; excluding the channel makes the bias appear in r_i at full amplitude (with a retrained classifier the practical effect is small; Section 6.6). Each observer is re-anchored only on channels the governor currently trusts, with weights that are slewed rather than switched (a step change in an anchor's weight would itself appear as a jump in other channels' residuals). Residual location and scale are learned on a clean commissioning window and then frozen: an adaptive scale is precisely the handle a slow injection would use to normalise itself away.

In the simulated testbed (Section 5.1) the process is a heated, sealed vessel with a bang-bang controller, and the observers combine Newtonian heating, the isochoric ideal-gas law $P \propto T$ and conservation of absolute humidity through the Magnus relation [39]:

$$e_s(T) = 610.94 \cdot \exp(17.625\ T / (T + 243.04)), \quad RH = 100\ e / e_s(T) \quad (2)$$

In the real-data embodiment (Section 5.2) there is no actuator; the expectation of a mote's temperature fuses the Magnus inversion of its own humidity at the neighbours' vapour pressure with the neighbours' mean temperature (inverse-variance weights fixed during commissioning), and the expectation of its humidity follows from its temperature and the neighbours' vapour pressure.

4.2 Residual-character features (E2)

A rate-matched injection and benign drift have, by construction, the same residual magnitude and slope. The features therefore describe *how* the residual evolves (Table 2). Their physical rationale is that a drifting transducer still observes the real process: its innovation statistics and its response to the modelled micro-dynamics are preserved, whereas a substituted stream carries the attacker's noise and does not follow the plant. The most legible example is excitation coherence,

$$\kappa_i(k) = -\,\mathrm{corr}_W(\,\Delta r_i,\ \Delta \hat{y}_i\,) \quad (3)$$

the negated correlation, over a window W, between residual increments and model-predicted increments: near zero for a sensor that tracks the real dynamics, positive for a stream whose increments do not. Forging it requires the attacker to reproduce the plant's micro-dynamics in real time — which is exactly the knowledge axis of the hierarchy in Section 3.2.

Table 2. Feature vector of the root-cause learner (22 features).

Group	Features	What separates the causes
Size	z_now, z_abs_max, r_mean_norm, r_std_norm, slope_norm, persistence, exceed_frac	Trigger context only; uninformative between rate-matched causes by design
Ageing envelope & onset	slope_ratio (vs. ρ_i), monotone_frac, corner, jump_max	Ageing is smooth, one-directional and inside ρ_i; an injection has an onset
Noise signature	acf_shift, var_ratio, flatness, hf_ratio, kurt_incr	Changes in the innovation statistics relative to commissioning
Dynamics	dyn_coherence (κ)	Does the stream still follow the modelled micro-dynamics?
Localisation (E4)	loc_score, loc_margin, loc_magnitude, loc_is_self, loc_is_process	Which hypothesis best explains the constraint-violation direction

4.3 Physics-signature localisation (E4)

A vector g(k) of physical-constraint violations is evaluated on the reported measurements and normalised by commissioning statistics (vessel: gas law, humidity conservation, open-loop dynamics, energy balance, regulation; indoor: Magnus consistency, spatial temperature, spatial humidity). For each hypothesis h — a unit bias on each channel, or an unmodelled process input — a signature s_h is obtained by perturbing the *model* at the commissioning operating point, so no labelled fault data are needed. Localisation reports the hypothesis that best explains the direction of the windowed mean violation $\bar{g}$, its share of the violation energy and its margin over the runner-up:

$$h^* = \operatorname{argmax}_h \left(s_h^T \bar{g} / \|\bar{g}\| \right)^2 \quad (4)$$

This answers a question orthogonal to drift versus poisoning — is the deviation on the instrument side or has the plant moved? — and it is what protects a genuine process event from being calibrated away. In the indoor embodiment, for example, local heating moves temperature and humidity together at constant vapour pressure and leaves the Magnus constraint intact, whereas a transducer bias violates it.

4.4 Root-cause learner (E5)

The feature vector feeds a four-class classifier. Training data are collected by running the pipeline with no classifier, so features at training time are exactly those at inference time; non-target channels are labelled NORMAL, which teaches the learner to stay quiet on collateral triggers. A thin physical override is applied in the hybrid embodiment: a confident process-side localisation is authoritative for PROCESS (where being wrong is a safety issue), and a PROCESS verdict is overruled when the violations localise cleanly to the channel itself. The learner is interchangeable; we evaluate a random forest [40] (default), gradient boosting, logistic regression, a multilayer perceptron and a training-free rule set. Because a fog node classifies one sample at a time, the forest is flattened into arrays and all trees are traversed in a single vectorised pass, numerically identical to the library implementation (asserted in the test suite) and about 200× faster for single samples.

4.5 Adaptation governance and differential response (E7–E8)

Each channel carries a state in {TRUSTED, WATCH, DRIFTING, QUARANTINED, RECOVERING}. A channel whose trigger fires leaves TRUSTED immediately, before any cause is decided, and from then on contributes reduced weight to the process model and nothing to the residual statistics. The response is cause-dependent:

- **DRIFT** → bounded, rate-limited, audited recalibration $c_i \leftarrow \text{clip}(c_i + \text{clip}(\gamma\, \hat{b}_i, \pm\delta), \pm c_{max})$, with $\hat{b}_i$ the robust residual bias; data keep flowing. The drift state is sticky while the correction is in force, because a residual held at zero by the correction is evidence that the correction works, not that the sensor recovered.
- **POISON** → quarantine: nothing is forwarded, every adaptive element is frozen, a security alert is raised, and the channel is held for a minimum period before it may recover.
- **PROCESS** → forward and alarm; calibration is forbidden.

Quarantine policy. A quarantined channel is not released by a drift verdict — otherwise an attacker could buy recalibration by steering one verdict — and the node acts on the governor's standing state rather than on the latest verdict, so a drift verdict on a quarantined channel never forwards data. By default the quarantine is *latched*: the channel returns only after the freeze hold and a clean run. We also evaluate an availability-oriented alternative, *re-attribution*, in which a quarantined channel moves to DRIFTING after the hold plus a sustained run of 200 consecutive drift verdicts; Section 6.4 shows why latching is the safer default.

Uncertain verdicts. A verdict below the confidence threshold (0.62) is escalated to a heavier back-end when one is configured. Without one, an uncertain POISON verdict holds that sample — it is neither forwarded nor admitted to any adaptive element — but it is recorded by the governor as UNKNOWN and cannot latch a quarantine; only a confident POISON verdict does. Uncertain DRIFT and PROCESS verdicts act locally, since their actions are bounded and reversible. This realises the rule that no irreversible action is taken on an ambiguous sample.

4.6 Cost

Per sample and channel, the nominal path is one model evaluation, a subtraction and a CUSUM update; features, localisation and classification run only on engaged channels, at a configurable re-evaluation interval (a fresh trigger always forces re-evaluation). Section 6.7 reports measured latency.

5. Experimental design

5.1 Simulated benchmark

The testbed is a heated vessel (thermal capacitance 5 kJ/K, loss 20 W/K, 400 W heater, bang-bang control at 30 °C with 0.3 °C hysteresis, sinusoidal ambient with a random-walk component) instrumented with internal temperature T_{in}, pressure P, humidity RH and ambient temperature T_{amb}, each with coloured AR(1) noise and quantisation. The fog node's model is deliberately mismatched (thermal capacitance +3 %, loss −2.5 %, heater +1 %, reference pressure +0.08 %, vapour pressure +1.5 %). Six scenarios are generated: clean, benign drift, rate-matched poisoning (A0), spike poisoning, replay poisoning and a process event (heater contactor welded on or failed open). Onset and rate are drawn per episode (onset in the second quarter of the episode; rate 0.8–2.5 × 10^{-3} units per sample), and the event polarity — drift or ramp direction, heater on or off — is balanced. Episodes are 4000 samples, of which the first 600 are a scored-out warm-up.

Splits. Models are trained on 6 seeds × 6 scenarios × 2 polarities (72 episodes, 509 513 labelled channel-samples) and evaluated on 30 disjoint seeds × 6 scenarios (180 episodes). Every reported number comes from the held-out bank.

5.2 Real data: Intel Berkeley Research Lab

The Intel Berkeley Research Lab deployment [21] recorded temperature, humidity, light and battery voltage from 54 Mica2Dot motes in an office between February and April 2004. We bin the ≈31 s epochs into 2-minute samples, interpolate gaps of up to 30 minutes, and use days 0–10, before battery failures make readings meaningless; 51 motes have ≤2 % missing data after interpolation. The clean signal — with its real micro-climate, occupancy and packet loss — is therefore real. Ground truth for drift and poisoning can only exist if events are injected, so they are injected into the real temperature streams (semi-synthetic evaluation): drift and A0 poisoning at 0.6–2.0 × 10^{-3} °C per sample, spikes, a replay of the previous day, and a genuine-process surrogate — a local heat source or draught that changes temperature by 1.5–4 °C and humidity exactly as psychrometrics requires at constant vapour pressure. Each mote uses its four nearest neighbours as physical references. **Motes are split at random into disjoint train and test halves**; every model is fitted on train motes only. Clean test episodes measure false alarms on unmodified real data.

5.3 Comparison systems

All learned baselines are trained on the same episodes with the same four labels and see at least the information PIRCAD sees — every raw channel plus the actuator command or neighbour aggregates

— but no physics (Table 3). Self-calibrating baselines were tuned to minimise their own correction error on benign drift (training seeds only), so that none can appear safe merely by not adapting.

Table 3. Comparison systems.

System	Type	Input	Output
Multi-scale 1D-CNN	Supervised deep learner; dilated convolutions over the last 128 samples and an 8× pooled 1024-sample context	All raw channels + command, standardised on commissioning	4 causes
Generic-feature RF	Random forest over multi-scale window statistics incl. rolling cross-channel correlations	As above	4 causes
Gen-1 FHAD	Rolling-mean detrending + CUSUM + windowed classifier (authors' earlier pipeline)	Target channel	4 causes
Isolation Forest [22], PCA-SPE, autoencoder	Unsupervised detectors on generic features / raw windows, 1 % clean FPR	All channels	anomaly flag
Kalman χ^2 [41]	Single closed-loop observer over all channels, windowed NIS test	All channels + command	anomaly flag
EWMA self-recalibration	EWMA reference; z-test-triggered absorption	Target channel	corrected value
Kalman bias estimator	Augmented state [T, b], same process model as PIRCAD	Target + command + ambient	corrected value
Page-Hinkley recalibration [23]	Detect persistent shift → recalibrate → restart	Target channel	corrected value

5.4 Metrics and statistics

Root-cause accuracy and **macro-F1** over the classes present in an episode are computed per sample after warm-up. Two directional errors are reported separately because they have opposite consequences: the **false-drift rate** (poisoned samples labelled drift — the attack is recalibrated into the reference) and the **false-poison rate** (drifting samples labelled poisoned — a serviceable sensor is discarded); the **process-masking rate** is the fraction of genuine process-event samples labelled as an instrument fault. Effect-level metrics measure what the system *does*: **poisoning containment** (poisoned samples not forwarded), **retraining protection** (1 − poisoned samples admitted to any adaptive element / poisoned samples) and **forwarded error** (mean absolute error between the forwarded, corrected value of the target channel and its true physical value over the event window, with withheld samples contributing zero). All metrics are computed per episode; we report means with 95 % bootstrap confidence intervals (4000 resamples) and compare systems with paired two-sided Wilcoxon signed-rank tests [42] matched by episode, with Holm correction [43] and rank-biserial effect sizes.

6. Results

6.1 In-distribution benchmark

Table 4. Root-cause attribution on 180 held-out simulated episodes (30 seeds × 6 scenarios). Mean [95 % bootstrap CI]; p: paired Wilcoxon test of per-episode macro-F1 against PIRCAD (random forest), Holm-corrected. ↓ lower is better.

System	Macro-F1	Accuracy	False-drift ↓	False-poison ↓	Process masked ↓	False alarm ↓	p
PIRCAD (random forest)	0.943 [0.932, 0.954]	0.941 [0.929, 0.953]	0.002	0.001	0.017	0.030	ref.
PIRCAD (gradient boosting)	0.962 [0.950, 0.972]	0.946 [0.933, 0.958]	0.001	0.072	0.022	0.031	$<10^{-4}$
PIRCAD (MLP)	0.863 [0.836, 0.889]	0.859 [0.827, 0.890]	0.001	0.393	0.010	0.112	$<10^{-4}$
PIRCAD (logistic regression)	0.673 [0.636, 0.711]	0.655 [0.612, 0.699]	0.033	0.758	0.021	0.368	$<10^{-4}$
PIRCAD (rules, no training)	0.643 [0.602, 0.683]	0.597 [0.556, 0.637]	0.080	0.625	0.433	0.211	$<10^{-4}$
Multi-scale 1D-CNN	0.938 [0.922, 0.954]	0.989 [0.988, 0.991]	0.005	0.001	0.001	0.004	0.274

System	Macro-F1	Accuracy	False-drift ↓	False-poison ↓	Process masked ↓	False alarm ↓	p
Generic-feature RF	0.974 [0.964, 0.982]	0.988 [0.986, 0.990]	0.000	0.001	0.000	0.001	$<10^{-4}$
Gen-1 FHAD	0.454 [0.402, 0.504]	0.514 [0.462, 0.564]	0.000	0.000	0.000	0.000	$<10^{-4}$

Table 4 addresses the first question: is a physics-anchored learner more accurate than a strong data-driven one when train and test come from the same distribution? It is not. PIRCAD reaches macro-F1 0.943 with a random forest and 0.962 with gradient boosting; the multi-scale 1D-CNN reaches 0.938 (not significantly different from PIRCAD-rf, $p = 0.27$) and the generic-feature forest 0.974, significantly higher. Both learned baselines see every raw channel and the actuator command, so in distribution they can learn the coupling the physics encodes. The authors' earlier pipeline (Gen-1), against which our prototype was originally compared, reaches 0.454: a comparison against it alone would overstate the contribution.

Three properties of PIRCAD's errors matter more than its mean. The two operationally dangerous confusions are near zero: poisoning read as drift (0.002) and drift read as poisoning (0.001). Its residual error is concentrated in benign drift, where the verdict is withheld until the ramp rises out of the noise (mean detection delay 69 samples), and in a false-alarm rate of 0.030 on clean data. And the back-end matters less than its calibration: gradient boosting and the random forest are within 0.02 of each other, the MLP reaches 0.86 because it issues confident poisoning verdicts early in benign drift, which the latched quarantine then holds (false-poison rate 0.39), and the linear model and the training-free rule set are far behind — the feature space is informative but not linearly separable.

Table 5. Detection only (anomalous vs. not), all scenarios pooled. Unsupervised detectors are thresholded at 1 % false alarms on clean training data.

System	Detection F1	Recall	False alarm
PIRCAD (random forest)	0.818	–	0.030
Multi-scale 1D-CNN	0.862	–	0.004
Isolation Forest	0.444	0.616	0.090
PCA-SPE	0.738	0.994	0.504
Autoencoder	0.760	0.935	0.138
Kalman χ^2	0.749	0.896	0.024

Table 5 places the unsupervised detectors. They flag most deviations but cannot name them, and none has a response that differs by cause; the Kalman χ^2 bad-data detector recalls 0.95 of rate-matched poisoning but would answer drift and poisoning identically. PIRCAD's own detection F1 is lower than the learned baselines' because it deliberately withholds a cause until the evidence supports one.

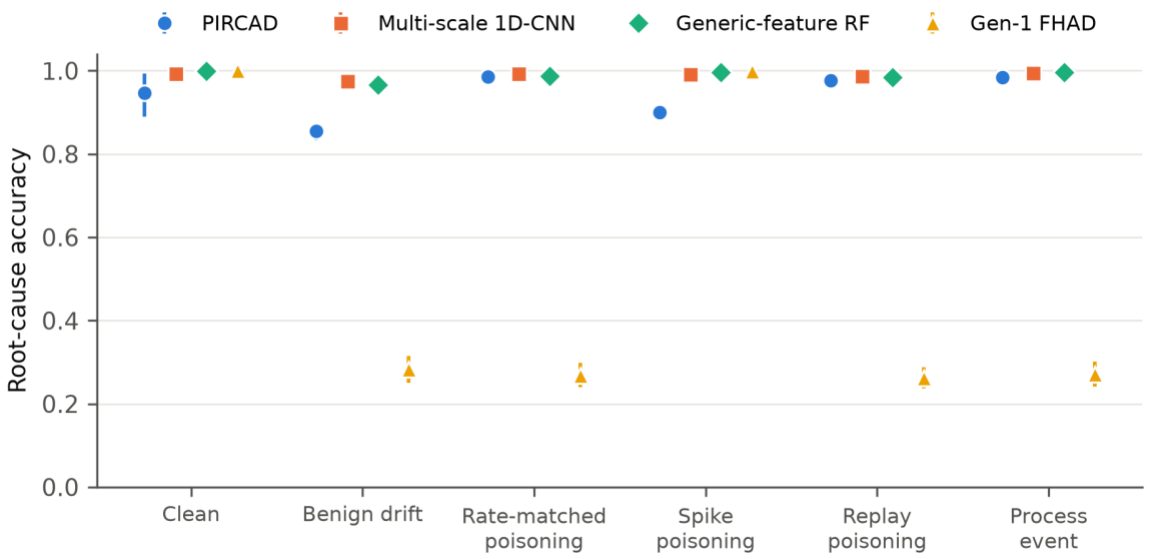


Fig. 2. Per-scenario root-cause accuracy on the held-out bank (mean, 95 % CI). PIRCAD's shortfall is concentrated in benign drift (detection delay) and spike poisoning; the learned baselines are uniformly strong in distribution, and Gen-1 can only name what a threshold on a rolling mean can see.

6.2 How much does the attacker need to know?

Table 6. Attacker-capability sweep (20 seeds per condition; models trained on A0 only). Recall is drift recall for the reference row and poisoning recall otherwise. Useful-data rate: fraction of event-window samples forwarded within 0.5 °C of the true value. Forwarded error: PIRCAD's mean absolute error over forwarded samples (°C).

Condition	Recall PIRCAD	Recall CNN	Recall RF	False-drift PIRCAD	Useful PIRCAD	Useful CNN	Useful RF	Fwd. error PIRCAD
Benign drift (reference)	0.808	0.967	0.954	–	0.99	0.03	0.04	0.17
A0 stale surrogate	0.989	0.992	0.982	0.000	0.01	0.01	0.02	0.10
A1 noise-matched	0.985	0.991	0.979	0.001	0.01	0.01	0.02	0.12
A2 fidelity 0.05	0.984	0.990	0.979	0.001	0.02	0.01	0.02	0.12
A2 fidelity 0.10	0.982	0.988	0.977	0.001	0.02	0.01	0.02	0.13
A2 fidelity 0.30	0.946	0.547	0.673	0.020	0.05	0.02	0.03	0.11
A3 exact state	0.357	0.000	0.755	0.532	0.61	0.03	0.04	0.18
A3 exact state + matched noise	0.003	0.002	0.002	0.833	0.99	0.03	0.04	0.18
A4 on-path (MITM)	0.002	0.001	0.003	0.831	0.99	0.03	0.04	0.17
A5 coordinated, stale	0.148	0.088	0.060	0.000	0.03	0.01	0.05	2.50
A5 coordinated, exact	0.000	0.002	0.000	0.010	0.13	0.04	0.05	2.19

Table 6 sweeps the threat hierarchy of Section 3.2 without retraining any model. Against attackers who know the live process only through a lagged surrogate (A0–A2 up to fidelity 0.10) every learned system recalls ≥ 0.98 of the poisoning; noise matching alone buys the attacker nothing. The systems separate as the surrogate sharpens. At fidelity 0.30 PIRCAD retains recall 0.95 against 0.67 (generic forest) and 0.55 (CNN), with the lowest false-drift rate; with the exact process state but white noise (A3) the generic forest is best (0.76 against 0.36 for PIRCAD) and the CNN collapses to zero. No method dominates: each has a different breaking point along the knowledge axis.

Once the attacker also matches the transducer's noise (A3) or adds the ramp to the genuine reading (A4), every system's poisoning recall falls to ≤ 0.003 — Proposition 1 observed: the attacked channel is statistically benign drift, and calling it drift is the only consistent verdict. What differs is the consequence. A label-only system can withhold what it flags but cannot correct it, because it has no estimate of the bias, so it delivers almost none of the affected channel: its useful-data rate on these

attacks — and on genuine drift — is 0.03–0.04. PIRCAD calls the A4 ramp drift, as it must, but corrects it against the physics: 99 % of the attacked channel is delivered within 0.5 °C of the truth, with a mean forwarded error of 0.17 °C, while the injected bias itself grows to several degrees. This is Proposition 2 observed: a mislabelled equivalent attack is neutralised in effect, which no amount of classification accuracy can achieve.

The coordinated attacker (A5) marks the limit of the method. By forging every physically coupled instrument from one fake temperature, it keeps the instrument-side constraints satisfied; PIRCAD's localisation then attributes the deviation to the process, and the node forwards it with a process alarm (forwarded error 2.5 °C). The alarm is raised, but the forged values are delivered. Defending against A5 requires an instrument the attacker does not control — an independent reference, an actuator-side measurement, or a cryptographically attested channel — which is outside the scope of a single fog node.

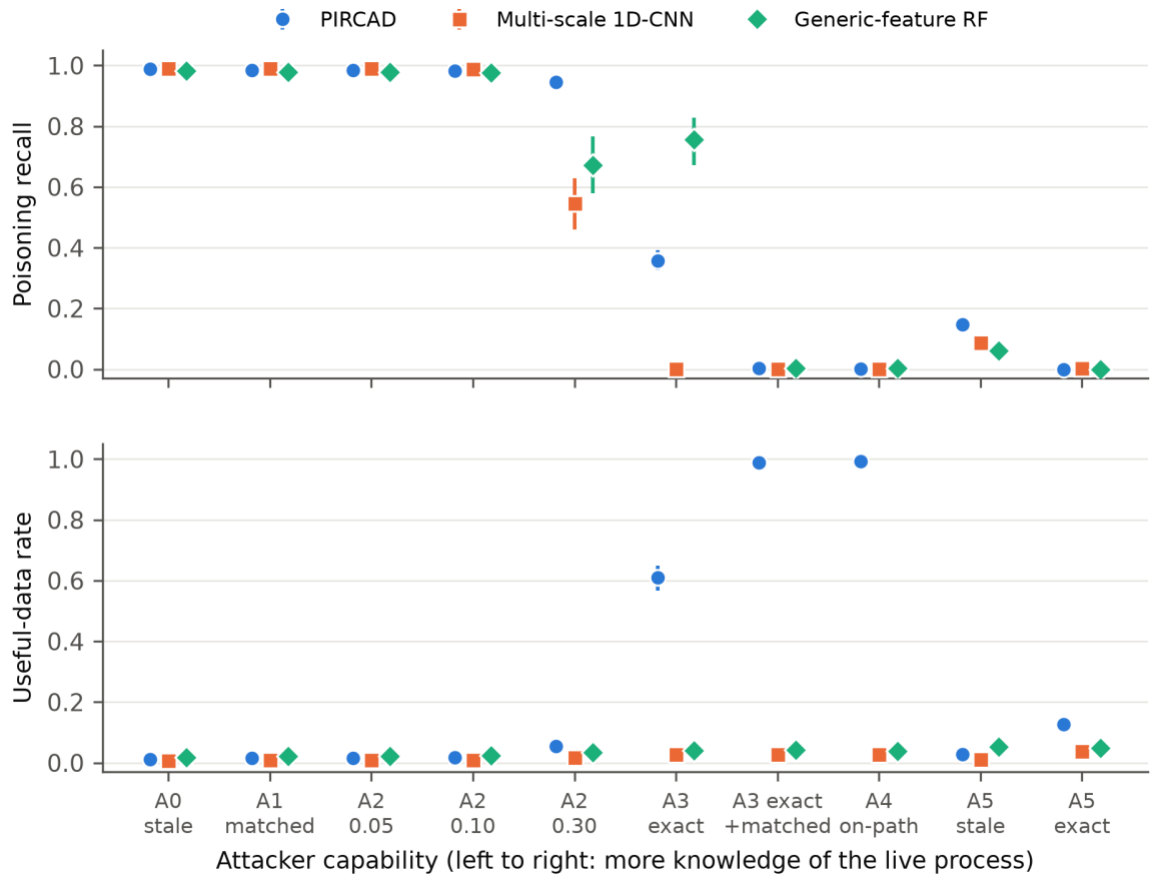


Fig. 3. Attacker-capability sweep. (a) Poisoning recall; (b) useful-data rate on the attacked channel. Left to right the attacker knows more about the live process. Where no system can label the attack (A3 with matched noise, A4), PIRCAD still delivers the channel corrected to the physical truth; label-only systems can only withhold it.

6.3 Robustness to distribution shift

Table 7. Macro-F1 under distribution shift, no retraining (10 seeds × 6 scenarios per condition). The process-model mismatch rows apply to PIRCAD only, since the learned baselines have no model to degrade. The last two columns are PIRCAD's false-drift and false-alarm rates.

Condition	PIRCAD	1D-CNN	Generic RF	False-drift PIRCAD	False alarm PIRCAD
Nominal	0.939	0.937	0.974	0.005	0.030
Process-model mismatch ×3	0.865	n/a	n/a	0.014	0.148
Process-model mismatch ×6	0.785	n/a	n/a	0.012	0.297

Condition	PIRCAD	1D-CNN	Generic RF	False-drift PIRCAD	False alarm PIRCAD
Plant dynamics −15 %	0.935	0.932	0.978	0.003	0.040
Plant dynamics +20 %	0.933	0.873	0.966	0.000	0.015
Sensor noise ×2	0.843	0.763	0.299	0.294	0.007
Sensor noise ×3	0.712	0.527	0.288	0.228	0.002
Ambient swing ×2.5	0.953	0.912	0.968	0.001	0.019
Set-point 35 °C	0.936	0.477	0.942	0.001	0.036
Rates 0.3–0.8 $\times 10^{-3}$	0.928	0.932	0.966	0.005	0.030
Rates 2.5–2.9 $\times 10^{-3}$	0.947	0.938	0.976	0.005	0.030

Table 7 trains every system once, on the nominal plant, and evaluates it on plants and sensors it has never seen. The data-driven learners are excellent until a shift touches what they learned, and then they fail abruptly. The generic-feature forest is unaffected by changed plant dynamics, ambient swing or set-point, but when the transducers become twice as noisy its macro-F1 falls to 0.299 with a false-alarm rate of 0.88: its features are scaled to the commissioning noise level, so the whole clean stream now looks anomalous. The CNN degrades under noise (0.527 at ×3) and collapses when the plant is merely regulated at a different set-point (0.477, false alarms 0.82) — a shift that changes nothing about the causes, only the operating point the network memorised.

PIRCAD degrades but does not collapse: its worst condition is sensor noise ×3 at macro-F1 0.712, against 0.477 (set-point 35 °C) for the CNN and 0.288 (sensor noise ×3) for the generic forest, and its mean over all conditions is 0.889 against 0.810 and 0.817. The residual is invariant to most of what changed — the set-point, the ambient and the plant dynamics enter the expectation as well as the measurement — so only shifts that alter the residual itself hurt. Two of those are real weaknesses. Under the strongest noise shift PIRCAD's false-drift rate rises to 0.23, because the noise-signature features that separate a substituted stream from a drifting one are calibrated to the commissioning noise; and an inaccurate process model (mismatch ×6) raises its false-alarm rate to 0.30. PIRCAD's robustness is therefore conditional on a process model of moderate accuracy, which is the price of judging a channel against physics rather than against its own past.

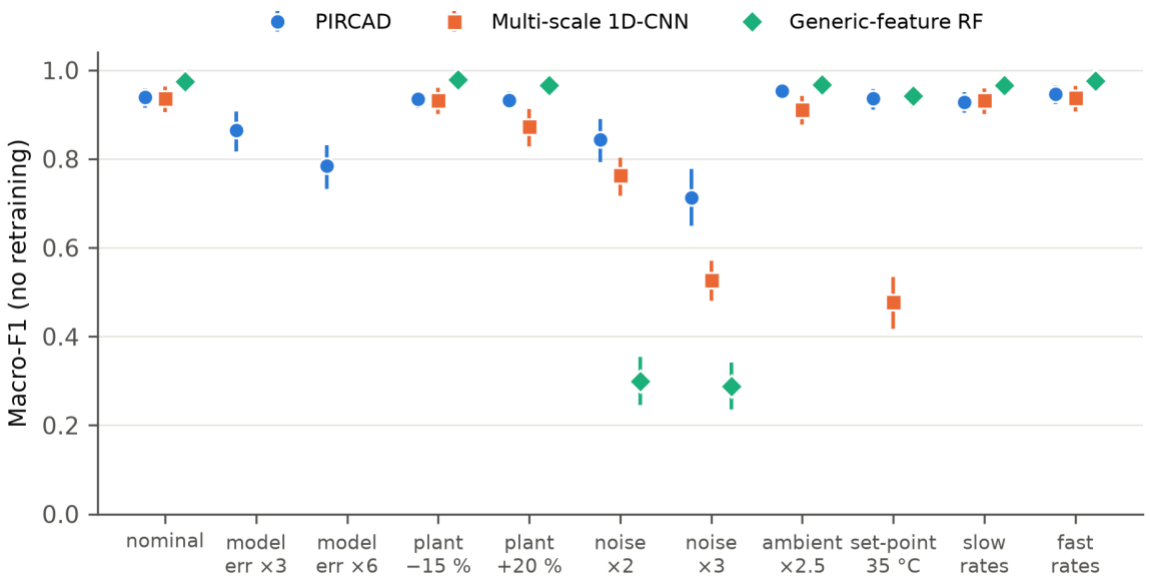


Fig. 4. Macro-F1 under distribution shift without retraining (mean, 95 % CI). Each data-driven learner has a condition under which it collapses; PIRCAD's worst case is bounded away from them. Mismatch conditions apply to PIRCAD only.

6.4 Does the safety function survive the attack?

Label accuracy is a proxy. The quantity that matters is whether the system's adaptive elements survive an attack well enough to report the next genuine fault. In each of 120 episodes (20 seeds × 3 attack rates × 2 attackers: A0, and a fidelity-0.30 noise-matched A2), a rate-matched injection biases the internal temperature downwards for 4500 samples; the attacker then withdraws and the heater contactor welds shut, a genuine over-temperature fault. A cloud-side alarm at 34 °C is applied to whatever the fog layer forwards. Each self-calibrating baseline was tuned on benign drift (training seeds) to do its own job well, and a benign-drift arm (60 episodes, no attack) measures that job.

Table 8. Attack-then-fault (120 episodes) and benign drift (60 episodes). Spurious trip: the forwarded value crossed 34 °C before the fault existed. Premature trip: it crossed after the fault began but ≥ 5 samples before the plant did (over-reading a real excursion; fails safe). Error at crossing: forwarded minus true temperature when the plant truly crossed 34 °C, mean [95 % CI]. The last column is the mean error of the forwarded, corrected value during benign drift.

System	Alarm fired	Spurious trip	Premature trip	Error at crossing (°C)	Poison forwarded	Drift correction error (°C)
PIRCAD	1.00	0.01	0.17	+0.18 [+0.09, +0.30]	0.021	0.41
PIRCAD auto re-attribution	1.00	0.27	0.19	+1.46 [+1.08, +1.88]	0.135	0.43
PIRCAD non-differential	1.00	0.72	0.13	+2.94 [+2.54, +3.34]	1.000	0.44
EWMA recalibration	0.00	0.33	0.00	-6.43 [-6.49, -6.38]	1.000	2.60
Kalman bias estimator	0.00	0.67	0.00	-4.72 [-4.75, -4.68]	1.000	0.14
Page-Hinkley recalibration	0.00	0.67	0.00	-3.69 [-3.71, -3.67]	1.000	0.10

Table 8 shows the consequence of absorbing an injection. None of the three self-calibrating baselines ever raises the alarm: each adopts part of the attacker's bias as its correction, and when the real fault arrives it under-reports by 3.7–6.4 °C. The tuned Kalman bias estimator is instructive: it uses the same process model as PIRCAD and is the best drift corrector in the table (0.14 °C), yet it fails the safety test, because a model-based bias estimator with no cause vocabulary must treat a slow injection exactly as it treats ageing. PIRCAD raises the alarm in 100 % of episodes, with an error of +0.18 °C at the crossing, a spurious-trip rate of 0.008, and 2.1 % of poisoned samples forwarded.

The two PIRCAD variants isolate where this comes from. With the response made non-differential — the same detector and classifier, but every detected deviation answered with recalibration — the alarm still fires, but 100 % of the poisoning is forwarded and the forwarded value over-reads by +2.94 °C at the crossing, with spurious trips in 72 % of episodes: detection without a cause-dependent response does not protect the safety function. With *auto re-attribution* instead of the latched quarantine, spurious trips rise to 0.27 and the crossing error to +1.46 °C, because the noise-matched attacker occasionally sustains enough drift verdicts to be re-attributed and partly corrected-in, while benign drift is corrected no better (0.43 against 0.41 °C). This is why latching is the default.

PIRCAD pays for this protection in drift-correction accuracy: its correction error during benign drift (0.41 °C) is larger than that of the tuned Kalman and Page-Hinkley correctors (0.14 and 0.10 °C), because its correction is rate-limited and is only engaged after the drift verdict. A deployment that faces no adversary should prefer those correctors; one that does should not use them unguarded.

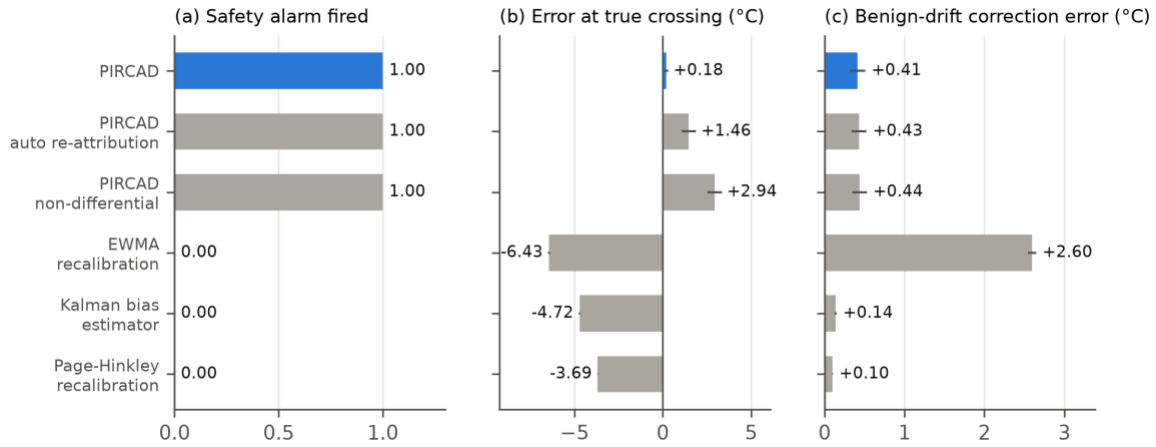


Fig. 5. Safety under adaptation poisoning. (a) Fraction of episodes in which the cloud-side 34 °C alarm fired for the genuine fault; (b) forwarded minus true temperature at the true crossing (negative = under-report, the dangerous direction); (c) correction error during benign drift. Means with 95 % CIs.

6.5 Real sensor data: Intel Berkeley Research Lab

Table 9. Intel Lab, 26 held-out motes × 6 scenarios × 2 replicates (312 episodes); models fitted on 25 disjoint train motes. Mean [95 % CI]; p: paired Wilcoxon on macro-F1 vs. PIRCAD (random forest), Holm-corrected. Harm: mean forwarded error on the injected channel (°C).

System	Macro-F1	Drift recall	Poison recall	Process recall	False-drift ↓	False alarm ↓	Harm ↓	p
PIRCAD (random forest)	0.712 [0.688, 0.735]	0.372	0.709	0.501	0.002	0.161	0.23	ref.
PIRCAD (gradient boosting)	0.664 [0.638, 0.688]	0.128	0.759	0.559	0.001	0.222	0.35	$<10^{-4}$
PIRCAD (rules, no training)	0.649 [0.620, 0.677]	0.017	0.843	0.303	0.001	0.177	0.07	$<10^{-4}$
Multi-scale 1D-CNN	0.854 [0.834, 0.874]	0.859	0.669	0.958	0.008	0.079	0.03	$<10^{-4}$
Generic-feature RF	0.845 [0.828, 0.862]	0.838	0.714	0.946	0.004	0.095	0.04	$<10^{-4}$
Gen-1 FHAD	0.355 [0.324, 0.387]	0.016	0.051	0.072	0.006	0.154	1.96	$<10^{-4}$

Table 9 is the least favourable result for PIRCAD in this paper, and we report it as such. On real streams with mote-disjoint splits, PIRCAD reaches macro-F1 0.712; the CNN and the generic forest reach 0.854 and 0.845, significantly higher. PIRCAD detects poisoning about as well as they do (recall 0.71 against 0.67 and 0.71) and never confuses it with drift (0.002), but it recognises real benign drift far less often (0.37) and raises more false alarms on clean real data (0.16). The reason is the expectation model. In the vessel the physics is exact up to a few per cent; in an office the leave-one-out expectation of a mote's

temperature combines a psychrometric relation that assumes spatially uniform vapour pressure with a spatial relation that ignores sunlight, vents and occupants, and its residual carries real micro-climate that the commissioning statistics cannot anticipate. The character features that separate a drifting transducer from a substituted stream are then measured on a noisier residual. The data-driven learners, trained and tested on the same building and the same injection generator, do not pay that price.

Table 10. Intel Lab under a transducer-noise shift (every mote given extra white noise; no retraining). Macro-F1 and false-alarm rate on held-out motes.

Condition	Macro-F1 PIRCAD	Macro-F1 Multi-scale 1D-CNN	Macro-F1 Generic-feature RF	False alarm PIRCAD	False alarm Multi-scale 1D-CNN	False alarm Generic-feature RF
No added noise	0.712	0.854	0.845	0.16	0.08	0.10
Noise +0.10 °C / +0.3 %RH	0.680	0.854	0.690	0.14	0.08	0.15
Noise +0.20 °C / +0.6 %RH	0.661	0.854	0.672	0.15	0.08	0.05

Table 10 repeats the decisive shift of Section 6.3 on the real deployment: the trained models are applied to test motes whose transducers have become noisier. At the strongest level macro-F1 falls by 0.051 for PIRCAD, 0.001 for the CNN and 0.173 for the generic-feature forest. The generic forest degrades to PIRCAD's level, as in simulation, but the CNN is essentially unaffected: on real data, whose own variability is large relative to the added noise, its standardised multi-scale input absorbs the shift. The robustness advantage PIRCAD showed in simulation therefore does **not** transfer to this real deployment under this shift, and we do not claim it does.

Two simulated findings do carry over, in weakened form. The on-path ramp (A4) again cannot be labelled — the learned baselines call it drift (poisoning recall 0.02 and 0.00); PIRCAD flags 0.35 of it, largely because its noisier real-data residual triggers poisoning verdicts on drift-like ramps too, so this is not evidence of separability. And only the system that corrects against an independent reference delivers a usable channel: PIRCAD forwards 34 % of the attacked channel and 35 % of a genuinely drifting one within 0.5 °C of the truth, against 5–7 % for the label-only learners — a sixfold difference, but far from the 99 % achieved in simulation, because the real-data correction is only as accurate as the real-data expectation.

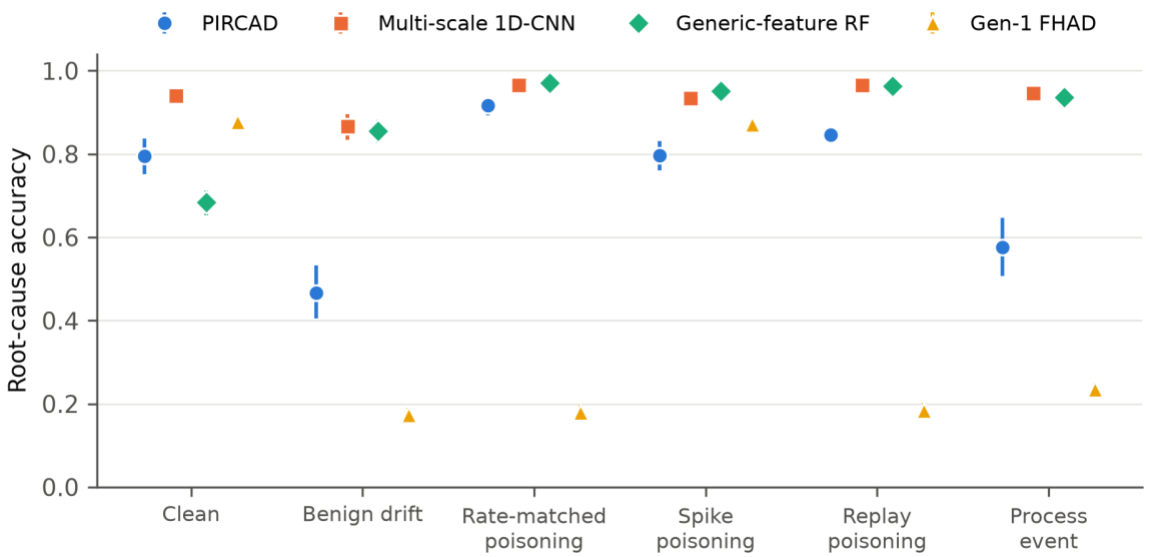


Fig. 6. Per-scenario root-cause accuracy on held-out Intel Lab motes (mean, 95 % CI). PIRCAD comes closest to the learned baselines on rate-matched poisoning and trails them elsewhere, most of all on real benign drift and local heat events.

6.6 Ablation and combined features

Table 11. Ablation on the 180-episode bank, one element removed per row; rows that change the features are retrained on a corpus collected under the same change. p: paired Wilcoxon on macro-F1 vs. the full system, Holm-corrected.

Configuration	Macro-F1	False-drift ↓	Containment ↑	Process masked ↓	Retraining protection ↑	Harm ↓	p
Full system	0.943 [0.932, 0.954]	0.002	0.973	0.017	0.998	0.05	ref.
− physics residual (rolling mean)	0.642 [0.597, 0.686]	0.019	0.512	0.352	0.551	0.41	$<10^{-4}$
− dedicated observers	0.930 [0.912, 0.947]	0.000	0.979	0.024	0.997	0.04	0.538
− cross-sensor consistency	0.937 [0.924, 0.949]	0.002	0.975	0.001	0.998	0.05	0.678
− excitation coherence	0.942 [0.931, 0.953]	0.001	0.971	0.018	0.998	0.05	0.538
− noise-signature features	0.956 [0.945, 0.966]	0.009	0.955	0.045	0.998	0.05	5.5×10^{-4}
− all character features	0.868 [0.841, 0.895]	0.008	0.935	0.130	0.998	0.02	$<10^{-4}$
− adaptation governor	0.947 [0.937, 0.957]	0.000	0.972	0.022	0.000	0.05	0.678
− differential response	0.908 [0.893, 0.923]	0.022	0.000	0.315	0.998	0.17	$<10^{-4}$

Table 11 removes one element at a time. The physics residual carries the result: replacing it by the channel's own rolling mean lowers macro-F1 from 0.943 to 0.642, cuts poisoning containment to 0.51, and lets 35 % of genuine process events be mislabelled as instrument faults. Removing the whole group of residual-character features costs 0.075 ($p < 10^{-4}$), whereas any single member is redundant with the others: excitation coherence alone changes nothing measurable (0.942, $p = 0.54$), and removing the noise-signature group even raises macro-F1 to 0.956 while raising the false-drift rate to 0.009 — a trade of label accuracy for the dangerous error.

Two results contradict hypotheses that motivated our initial design. Replacing the dedicated observers by one shared closed-loop estimator does not significantly change macro-F1 (0.930, $p = 0.54$) once the classifier is retrained on the changed features: the bias is attenuated in a shared estimator's residual, but the trigger and the remaining features still see it. And removing the cross-sensor localisation leaves macro-F1 unchanged (0.937, $p = 0.68$); in this plant the residual character alone suffices to recognise the process event. We therefore do not claim either element as necessary for classification in

distribution. The leave-one-out structure is kept because Proposition 2 requires an expectation that excludes the channel; in this plant a shared estimator happens to draw most of its weight from the other instruments, so the difference is small here but is not guaranteed in general.

The two governance rows act on effect, not on labels, which is why a label-only ablation would rank them as worthless. Without the adaptation governor, macro-F1 is unchanged but retraining protection falls from 0.998 to 0.000: every poisoned sample reaches an adaptive element. Without the differential response, containment falls from 0.973 to 0.000 and process masking rises from 0.017 to 0.315.

Table 12. Physics features, generic features, and both in one learner (macro-F1). The combined learner keeps PIRCAD's governance and response.

Condition	PIRCAD (physics)	Generic-feature RF	PIRCAD (physics + generic)	False alarm, combined
Nominal (180-episode bank)	0.943	0.974	0.976	0.001
Plant dynamics +20 %	0.933	0.966	0.947	0.001
Sensor noise ×2	0.843	0.299	0.624	0.001
Sensor noise ×3	0.712	0.288	0.525	0.001

Table 12 asks whether the two feature families are complementary. In distribution the combined learner reaches 0.976, closing the gap to the generic forest; under the strongest noise shift it reaches 0.525, so it inherits the fragility of the generic features. Adding data-driven features to a physics-anchored learner is therefore a trade-off along the same axis as the comparison itself, not a free improvement.

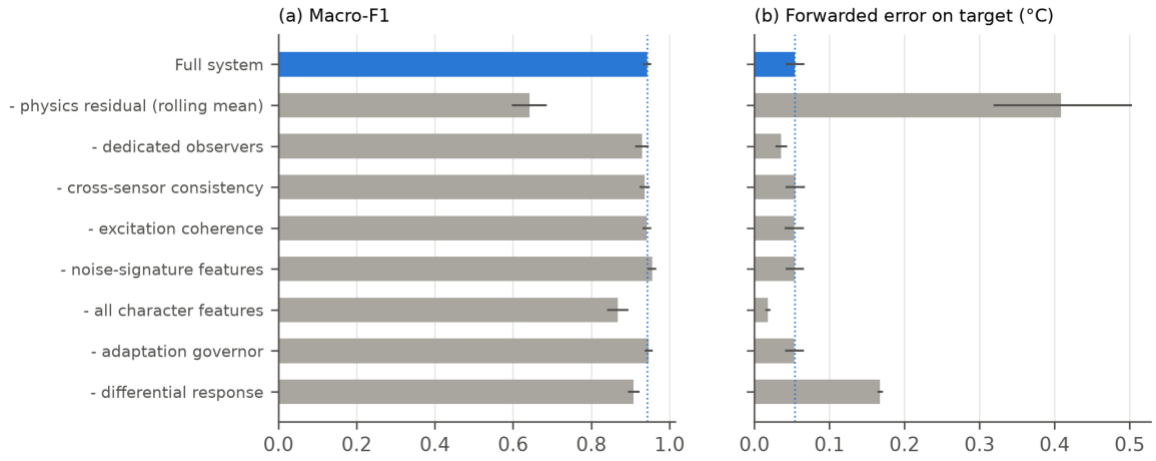


Fig. 7. Ablation (mean, 95 % CI). (a) Macro-F1; (b) forwarded error on the target channel. The dotted line marks the full system.

6.7 Computational cost

Over the 180 held-out episodes, PIRCAD processes a four-channel frame in a median of 0.42 ms (95th percentile 0.68 ms) in single-threaded Python on an Apple-silicon laptop, and the expensive stages run on 31 % of channel-samples. The single-sample forest evaluation of Section 4.4 is what makes this feasible: the library implementation spends about 16 ms per single-sample call on per-tree task dispatch, which made a straightforward implementation 30× slower with identical outputs. The two-channel edge kernel runs in closed form on an ESP32-class microcontroller (Section 6.8).

6.8 Constrained edge deployment

The same architecture was reduced to closed form for an ESP32 DevKit with a single DHT22, whose temperature and humidity transducers supply the cross-channel redundancy through the Magnus relation; the decision kernel (about 4 µs per sample, compiled C++) runs in the Wokwi simulator and is unchanged from our earlier prototype. With no learned model on the node, it recalls 88.1 % and 87.5 % of linear and exponential drift, 100 % of step and spike injections and 93.1 % of a drift-shaped ramp above the ageing envelope, and admits under 1 °C of a 6 °C step or a stealth ramp into its reference, against 5.8–6.8 °C for a rolling-mean baseline. A ramp matched to the benign drift rate is not separable with two channels at one node — the regression suite asserts that it fails — which is the single-node instance of Proposition 1. The node has not yet been run on physical silicon.

7. Discussion

7.1 What the evidence supports

The results support a narrower and, we think, more useful claim than the one we started with. PIRCAD is **not** a more accurate classifier than a strong data-driven learner trained on the same distribution: in simulation the CNN and the generic forest match or beat it, and on real data they beat it clearly. What a physics-anchored, cause-aware learner offers is different in kind:

- **Bounded worst case under shift — in simulation.** Every data-driven learner had a simulated shift under which it collapsed (worst macro-F1 0.29 and 0.48); PIRCAD's worst case was 0.71, provided its process model is reasonably accurate. On the real deployment the CNN was unaffected by the noise shift we could apply, so this advantage is not yet established on real data.
- **The ability to act on a verdict.** Correcting a drifting sensor requires an estimate of its bias against something it cannot influence. A label-only system can only withhold what it flags; on benign drift PIRCAD delivered 99 % of the channel within tolerance against 4 % for the learned baselines in simulation, and 35 % against 7 % on real data.
- **Harm bounded where labels are impossible.** For attacks that are observationally equivalent to drift (Proposition 1), no system labels them, but physics-anchored correction neutralises them in effect (Proposition 2, Table 6).
- **A safety function that survives poisoning.** Against tuned self-calibrating estimators that never raised the alarm, PIRCAD raised it in 100 % of attack-then-fault episodes.

A practitioner should therefore read the comparison as complementary rather than competitive. Where the deployment is stable and labelled data are plentiful, a data-driven learner is the better *classifier*; it is the governance — what may learn from the data, and how a verdict is acted on — that needs an independent reference. The combined-feature experiment (Section 6.6) shows the classifier back-end can be swapped without touching the governance, which is where we expect the most productive follow-up work.

7.2 Three defects exposed by the stronger evaluation

Our earlier prototype of this system was evaluated on four seeds against its own predecessor. Re-evaluating it at the scale reported here exposed three defects, each invisible to the original benchmark:

- **A polarity shortcut.** Every drift and attack ramp in the original training bank rose, and every process event heated. The classifier learned the sign: a falling drift was called normal, then a process event, and was never corrected. A first fix that drew the polarity at random happened, with six

training seeds, to make every training drift fall — and the model learned that rising drift is normal. Training on both polarities of every episode removed the shortcut.

- **Quarantine lock-in.** One transient poisoning verdict at the onset of genuine drift latched quarantine for the rest of the run, while the response still followed the latest (drift) verdict, so uncorrected data were forwarded from a channel the governor believed quarantined. Containment now follows the governor's standing state.
- **Irreversible action on uncertain verdicts.** With no escalation back-end the fallback forced an uncertain verdict to full confidence, so a 0.3-confidence poisoning guess latched quarantine. On real data this turned a classifier that mostly called drift correctly into one that reported poisoning for most of each drift episode. Uncertain poisoning verdicts now hold the sample without latching.

All three made the system look better or worse than it was in ways that a label-accuracy table on a small, one-sided bank cannot reveal. We report them because they are the most transferable lesson of this work for anyone evaluating an adaptive anomaly pipeline: balance the polarity of every event class, evaluate on effect as well as labels, and test the governance logic with episodes long enough for its state machine to matter.

7.3 Limitations and threats to validity

- **Simulation.** The decisive comparisons use a simulated plant whose physics is known up to a few per cent. The real-data study shows that a weaker expectation model costs PIRCAD accuracy; the method's benefits are only as good as its reference.
- **Semi-synthetic real data.** The Intel streams are real, but drift, poisoning and the local heat events are injected, by the same generator family used for training; the data-driven baselines benefit from that match. No public dataset provides labelled real ageing alongside real attacks.
- **Attackers.** The hierarchy covers knowledge of the process and the noise, on-path manipulation and coordinated forgery, but not an attacker who adapts to the classifier's gradients or controls the actuator commands.
- **Coordinated forgery (A5).** Forging every coupled instrument consistently defeats the constraint attribution; the node raises a process alarm but forwards the forged values.
- **Sensitivity.** PIRCAD's false-drift rate rises under strong transducer-noise shift, and its false-alarm rate under a poor process model.
- **Baselines.** The learned baselines were not tuned beyond sensible defaults; the self-calibrating baselines were tuned on benign drift only. A stronger attack-aware baseline (e.g. robust drift detection [20]) was not re-implemented.
- **Hardware.** The edge node has been run in simulation only; energy and memory on physical silicon are not reported.

8. Conclusion

Self-calibrating IoT pipelines can be poisoned by injections shaped like benign transducer ageing, and the harm outlasts the attack. We framed the defence as cause-aware learning whose verdicts govern every adaptive element, stated the limit of what any detector can separate, and showed that judging each channel against a physics expectation it cannot influence bounds the harm even beyond that limit. A statistically powered evaluation against deep and feature-based learners, unsupervised detectors and tuned self-calibrating estimators, including a real-data study, supports a precise conclusion: physics anchoring does not buy in-distribution accuracy, and on real data a well-trained CNN remains the better classifier; what it buys is the ability to correct rather than discard, harm that stays bounded where labels are impossible, a downstream safety function that survives poisoning, and — in simulation — a bounded

worst case under shift. The most promising direction is to combine the governance studied here with learned expectation models whose reference remains independent of the channel being judged, so that the robustness of the physics can be carried to deployments where exact physics is not available.

CRediT authorship contribution statement

Pushkal Gupta: Conceptualization (lead), Methodology (lead), Software (lead), Formal analysis (lead), Investigation (lead), Data curation (lead), Validation (lead), Visualization (lead), Writing – original draft (lead), Writing – review & editing (lead), Project administration (lead). **Shridhar M:** Software (supporting), Validation (supporting), Investigation (supporting), Writing – review & editing (supporting). **Vanka Kaushik:** Software (supporting), Validation (supporting), Data curation (supporting), Writing – review & editing (supporting). **Syamasudha Veeragandham:** Supervision (lead), Resources (lead), Conceptualization (supporting), Writing – review & editing (supporting).

P. Gupta conceived the drift-versus-poisoning attribution problem and the physics-anchored formulation, designed the threat model and the two propositions, implemented the fog pipeline, the attack and scenario generators, all comparison systems and the edge kernel, designed and ran every experiment, carried out the statistical analysis and the Intel Lab study, identified and fixed the three defects reported in Section 7.2, produced all figures and wrote the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The Intel Berkeley Research Lab data are publicly available [21]. All code — the fog pipeline, the scenario and attack generators, every comparison system, the experiment scripts that produce each table and figure, and the regression tests — is available at https://github.com/Pushkal-Gupta/Sensor-Drift-and-Data-Poisoning-in-IoT-Sensor-Networks.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with software development, the execution of the experiments, and drafting and editing of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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